MIL-D-17084E(SH)
22 October 1987
SUPERSEDING
MIL-D-17084D(SHIPS)
21 February 1963
(See 6.6)

MILITARY SPECIFICATION

DUMMY TRANSMITTERS, SHIPS' DISTANCE AND SPEED

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dummy log equipment for use on Naval ships for the purpose of providing synchro transmission of ships' speed and distance at a rate determined by manual control when the ship's underwater log system in inoperative.

1.2 Classification. Dummy log equipment shall be of the following types as specified (see 6.1):

Type I - 0 to 40 knots.
Type II - 0 to 25 knots.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic, Interior Communication, and Navigation Equipment Naval Ship and Shore, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

2.1.2 Government drawing. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

9000-S6504-73687 - Dial Markings for Interior Communication and Indicating Systems.

(Copies of specifications and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Description. The dummy log equipment shall be constructed to transmit indications of ships' speed and distance traveled by means of synchro transmission. The equipment shall consist of the following components:

(a) Dummy log speed transmitter.
(b) Dummy log distance transmitter.

3.3 General features. The equipment shall be in accordance with the following requirements of MIL-E-16400 in addition to the requirements specified herein.

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Iron and steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Gaskets
Dials and pointers (see 3.4.1.2)
Dial sizes
Locking devices
Washers
Ball bearings

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Parts - electrical - general
Use of nonstandard parts
Requirements for semiconductor devices
Capacitors
Variable resistors
Transformers
Relays
Synchros (see 3.4.1 and 3.4.2)
Synchro capacitors
Synchro electrical zero and equipment mechanical zero
Servo motors
Electrical tapes
Dial illumination lamps
Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuse clips
Metallic rectifiers
Printed wiring and circuits
Enclosures - general
Enclosure - accessibility
Enclosure - degree of (see 3.4.1.1 and 3.4.2.1)
Enclosure - mounting
Stiffening grooves
Minimum sheet metal thicknesses
Through belting
Cable entrance
Cable entrance plates
Ventilation
Size (for submarines and surface ships)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity
Accelerated life
Shock, vibration and inclination (vibration test for vital equipment required)
Features electrical
Primary power supply circuits
Power supply tolerances
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (test cable required)
Terminal boards connectors and terminals
Synchro connections and markings
Wiring
Color coding

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Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise
Structureborne noise (grade C)
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Drawings - manufacturing
Bill of materials
Interchangeability and standardization
Manuals (type I)
Repair parts
Repair parts (electronic)
Designation and marking (serial numbers are required)
Reports
Item names and nomenclature
Workmanship and general examination

3.4 Detail requirements.

3.4.1 Dummy log speed transmitter. The mechanism of the transmitter shall consist of a switch to control the positioning motor for the speed transmitting synchro in the dummy log distance transmitter and a nonilluminated dial with a pointer positioned by a type 23TR6 synchro which receives its signal from the speed transmitting synchro in the dummy log distance transmitter.

3.4.1.1 The speed transmitter shall be housed in a dripproof case, and shall be not larger than 7 inches wide by 9 inches high, by 6 inches deep. The case shall be constructed for panel mounting, shall be as compact and lightweight as practicable, and shall not exceed 15 pounds.

3.4.1.2 The dial of the speed transmitter shall be in accordance with Drawing 9000-S6504-73687. The markings shall be black on a dull white background.

3.4.2 Dummy log distance transmitter. The transmitter shall transmit both speed and distance by synchro transmission. It shall contain a type 37TX6 synchro to transmit speed at a uniform rate of 25 or 40 knots for 360 degrees of rotation, and a type 31TX6 synchro to transmit distance at the rate of 360 revolutions per nautical mile. The speed transmitting synchro shall be positioned by a motor controlled by the switch in the speed transmitter; the distance transmitter shall be positioned by a motor controlled by a follow-up mechanism attached to the output of an integrating device which integrates the speed set into the speed-transmitting synchro to distance. The distance transmitting synchro shall not be driven directly from the output of the integrating table. Devices that position the type 37TX6 and type 31TX6 synchros shall have 20 and 15 inch-ounces of torque, respectively.

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3.4.2.1 The distance transmitter shall be housed in a dripproof case, shall be not larger than 20 inches wide by 20 inches high by 10 inches deep, and shall weigh not more than 90 pounds. It shall be constructed for bulkhead mounting.

3.4.2.2 Accuracy. For a particular setting of speed on the speed transmitter, the distance transmitter shall transmit speed and distance within the following tolerances:

Speed - The error between the indicator in the speed transmitter and the transmitting synchro in the distance transmitter shall not exceed the normal synchro transmission error.
Distance - plus or minus 1/2 of 1 percent of the distance proportional to full scale speed.

3.4.3 Speed of response. The equipment shall be so constructed that speed changes can be made at the rate of 0 to full scale in 2 minutes plus or minus 30 seconds.

3.4.4 Distance indications shall not lag any corresponding speed indication in excess of the tolerances specified herein, by more than 15 seconds.

3.4.5 Continuous operation. The equipment shall be constructed for continuous operation.

3.5 On board repair parts. The following on board repair parts shall be supplied:

Name	Quantity
Motor starting capacitor	1
Electrical contact brushes (excluding synchro brushes)	1 set of each type and size.
Patch cables	1 of each for the speed transmitter and for the distance transmitter.
Springs (The failure of any one or more of these springs would cause the equipment to be unreliable or unoperative)	1 of each type and size.
Bearings	1 of each size and type except where bearings are replaced in sets. Under such circumstances 1 set of each size and type shall be supplied.
Special screws (screws which are not available from normal stocks due to size and/or shape)	1 box

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests listed in table I, which shall be performed in the order listed.

TABLE I. First article inspection.

Tests	Requirement	Test
General examination	3.3	MIL-E-16400
Operating	3.3	4.5.1
Accuracy and speed of response	3.4.2.2, 3.4.3	4.5.2
Supply line voltage and frequency variation	3.3	MIL-E-16400
Radio frequency noise	3.3	MIL-E-16400
Airborne noise	3.3	MIL-E-16400
Structureborne noise	3.3	MIL-E-16400
Temperature and humidity	3.3	MIL-E-16400
Inclination	3.3	MIL-E-16400
Accelerated life	3.3	MIL-E-16400
Dielectric strength	3.3	MIL-E-16400
Insulation resistance	3.3	MIL-E-16400
Enclosure	3.3	MIL-E-16400
Vibration	3.3	MIL-E-16400
Shock	3.3	MIL-E-16400

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The results of each examination and test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the first article samples and on future production units.

4.4 Quality conformance inspection. The equipments selectred in accordance with 4.4.1 shall be subjected to the tests listed in table II. Tests shall be performed in the order listed.

TABLE II. Quality conformance inspection.

Inspection	Requirement	Test
Group A		
General examination	3.3	MIL-E-16400
Operating	3.3	4.5.1
Accuracy and speed of response	3.4.2.2, 3.4.3	4.5.2
Dielectric strength	3.3	MIL-E-16400
Insulation resistance	3.3	MIL-E-16400
Group B		
Supply line voltage and frequency variation	3.3	MIL-E-16400
Airborne noise	3.3	MIL-E-16400
Group C		
Structureborne noise	3.3	MIL-E-16400
Radio frequency noise	3.3	MIL-E-16400
Temperature and humidity	3.3	MIL-E-16400
Inclination	3.3	MIL-E-16400
Accelerated life	3.3	MIL-E-16400
Enclosure	3.3	MIL-E-16400
Vibration	3.3	MIL-E-16400
Shock	3.3	MIL-E-16400

4.4.1 Sampling for quality conformance inspection.

4.4.1.1 Inspection lot. All equipment offered for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.4.1.2 Group A and B. All equipment covered on the contract or order shall be subjected to the group A and B examination and tests listed in table II. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any examination or test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

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4.4.1.3 Sampling for group C tests. Group C tests will be required by the Command or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests listed in table II. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.5 Test procedures.

4.5.1 Operating. The equipment shall be operated for a minimum of 60 minutes, during which time it shall be oscillated continually over the entire range.

4.5.2 Accuracy and speed of response. The equipment shall be checked for the following:

(a) Accuracy at a minimum of five points (see 3.4.2.2).
(b) Speed of response before and after the 60-minute oscillation period (see 3.4.3).

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early equipment installation and for storage of on board repair parts.

5.1.1 Dummy transmitter.

5.1.1.1 Preservation and packaging. Preservation and packaging shall afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the contractor's commercial practice when such meets these requirements.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice when such meets these requirements.

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5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, National stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 On board repair parts. On board repair parts shall be preserved and packaged level A, packed level C, and marked in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation and packaging, packing, and marking for shipment shall be as specified (see 6.2).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment which may be required when acquisition is made:

5.2.1.1 Preservation and packaging, packing, and marking. The equipment and accessories, repair parts, and technical publications shall be preserved and packaged level A or C, packed level A or B as specified, and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. Dummy log equipment is installed on ships which have installed equipment requiring synchro speed or distance data. It is used when it is necessary to operate with the sensing element retracted, when the log is inoperative, and when it is desired to train on fire control or navigation equipment which uses ships' speed and distance synchro data.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of manuals required.
(d) Inventory control point for stock repair parts (see MIL-P-15137).
(e) Whether unit stock number is to be entered on the identification plate (see MIL-E-16400).
(f) Preservation and packaging, packing, and marking if other than as specified in 5.1 (see 5.2).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Log equipment
Speed
Synchro transmission

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6320-N026)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-17084E(SH)	DUMMY TRANSMITTERS, SHIPS' DISTANCE AND SPEED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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